

1. Create a table listing and describing the main types of hackers (such as white hat, black hat, grey hat, script kiddie, hacktivist, and state-sponsored) and give one real-world example or news headline for each type.

Type of Hacker	Description	Real-world example
White Hat	Ethical hackers who have permission to test systems, find vulnerabilities and report them so they can be fixed.	Google Vulnerability Reward Program – Security researchers report vulnerabilities to Google and can receive rewards.
Black Hat	Hackers who access systems without permission for theft, fraud, disruption or other malicious purposes.	Bangladesh Bank cyber heist – Lazarus/related North Korean operators used stolen credentials and malware in an attempt to steal hundreds of millions of dollars.
Grey Hat	Hackers who may find or test vulnerabilities without permission but do not necessarily intend to cause serious harm. They may disclose the vulnerability afterward.	Unauthorized vulnerability discovery – a researcher might test a public website without permission, discover a security flaw and then notify the company. The activity can still be unauthorized even if the intention is helpful.
Script Kiddie	An inexperienced attacker who uses existing hacking tools or scripts created by others rather than developing their own techniques.	DDos-for-hire incidents – less-skilled attackers have historically used ready-made tools /services to launch attacks without understanding the underlying technology.

Hacktivist	A hacker or group that uses cyberattacks to promote a political, social or ideological cause.	Anonymous – The hacktivist collective has been associated with attacks and online campaigns against organizations for political or social reasons.
State-Sponsored Hacker	A hacker or group supported, directed or controlled by a government often for espionage, disruption, military or financial objectives.	Lazarus Group – The US Treasury identifies Lazarus, Bluenoroff and Andariel as North Korean state-sponsored groups. Lazarus has also linked to the 2014 Sony Pictures attack.

2. Pick any popular app you use (Instagram, Paytm, Zomato, etc.) and describe a scenario for each type of hacker: how would a white hat, black hat, and grey hat hacker interact with this app? Write 2-3 sentences per scenario.

Example of Instagram

White Hat Hacker:

- A white hat hacker may test Instagram's security with permission from Meta to find vulnerabilities. If they discover a flaw, they report it responsibly so the company can fix it before criminals exploit it.

Black Hat Hacker:

- A black hat hacker may try to steal Instagram user's passwords or personal information through phishing or other illegal methods. They could then use compromised accounts for fraud, spam or further attacks.

Grey Hat Hacker:

- A grey hat hacker might discover a security weakness in Instagram without having permission to test it. They may report the vulnerability afterward but accessing or testing systems without authorization can still violate rules or laws.

3. Given the following user actions, classify each as white hat, black hat, grey hat, or other: reporting a bug to the company, stealing user data, testing your own account for vulnerabilities, defacing a website for a political cause, and participating in a government cyber operation.

User Action	Classification	Reason
Reporting a bug to the company	White Hat	The person responsibly reports a security weakness so the company can fix it.
Stealing User Data	Black Hat	The person accesses or takes data without permission for malicious or illegal activities.
Testing your own account for vulnerabilities	White Hat	Testing an account you own with authorization is ethical security testing.
Defacing a website for a political cause	Hacktivist	This is typically hacktivism because the attack is motivated by a political or social cause.
Participating in a government cyber operation	State-Sponsored	Cyber operations conducted on behalf of a government are generally classified as state-sponsored activity.

4. Interview-style: Imagine you are explaining to your non-technical friend the difference between a white hat and a black hat hacker using a cricket match analogy. Write your explanation in 4-5 sentences.

- Imagine a cricket match where a white hat hacker is like a player who follows the rules and helps the team to find weaknesses so they can improve. A black hat hacker is like a player who cheats, break the rules and tries to harm the opposing team for personal gain. The white hat reports problems to the team or organizers while the black hat secretly exploits them. In cybersecurity, white hats protect systems while black hats steal data, cause damage or gain unauthorized access. So, simply white hats use their skills to protect while black hats use them to attack.